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N+ NETWORK FUNDAMENTAL AND BUILDING NETWORKS

SWITCH AND ROUTER CONFIGRATION BASIC THEORY

DAY:-3

TASKS:1

Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

RESULT

1. Opened Cisco Packet Tracer.
2. Added one 2960 Switch to the workspace.
3. Clicked on the switch and opened the CLI.
4. Entered the following commands:
   - enable

   configure terminal

   - vlan 10

   name Sales

   - vlan 20

   name Support

   - exit

   The VLANs Sales (10) and Support (20) were created successfully.

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TASKS:2

Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Use the interface range command to select multiple ports efficiently.

RESULT

ASSIGN PORTS TO VLAN 10(SALES)

enable

configure terminal

interface range fa0/1 - 3

switchport mode access

switchport access vlan 10

exit

ASSIGN PORTS TO VLAN 20(SUPPORT)

interface range fa0/4 - 5

switchport mode access

switchport access vlan 20

exit

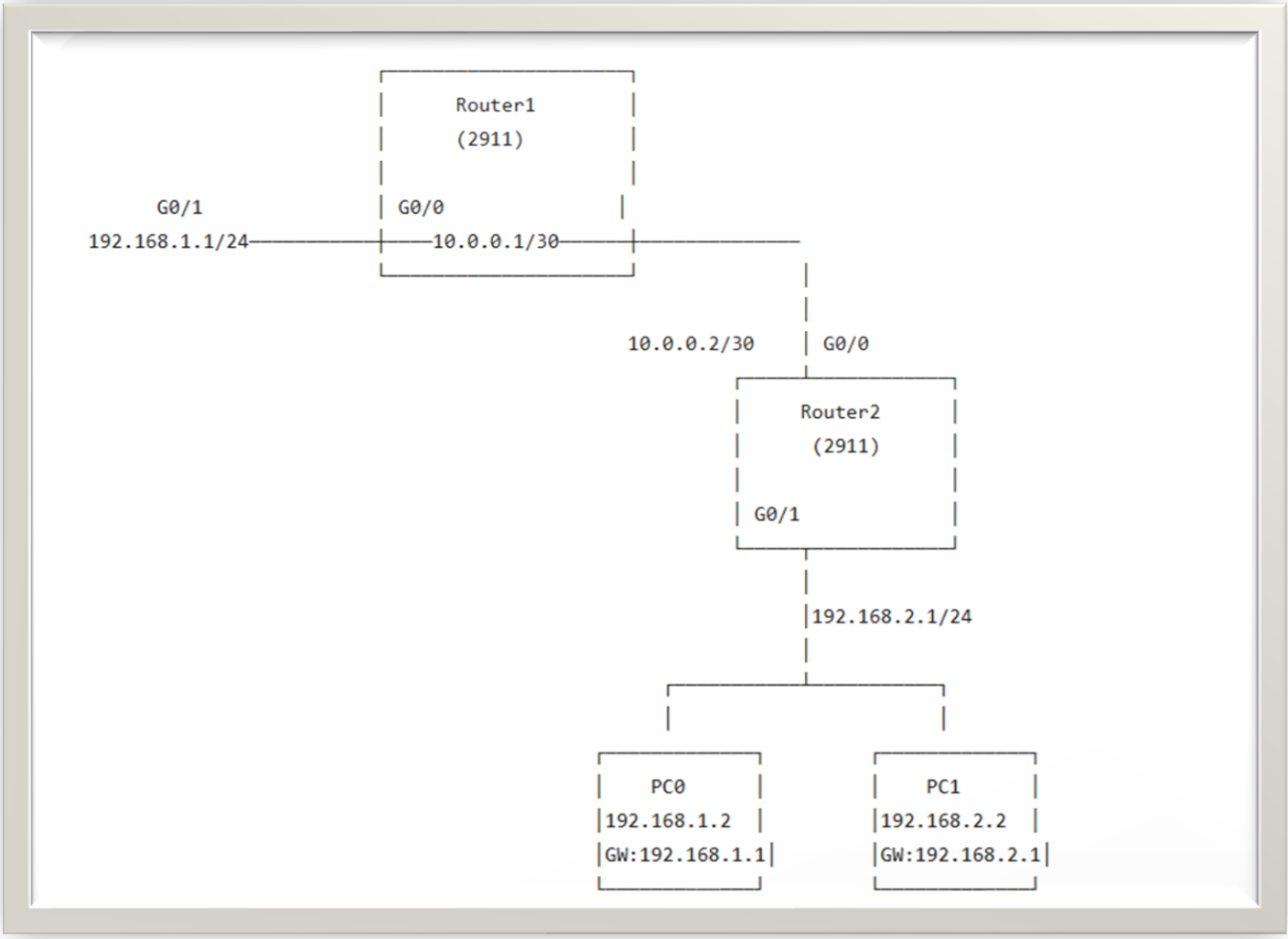
VERIFY VLAN: show vlan brief

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TASKS:3

Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

RESULT



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TASKS:4

Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.Think about router-on-a-stick or Layer 3 switch concepts.

RESULT

OBSERVATION:

A PC in the Sales VLAN (VLAN 10) tried to ping a PC in the Support VLAN (VLAN 20).

The ping failed because VLAN 10 and VLAN 20 are separate broadcast domains. A Layer 2 switch cannot forward traffic between different VLANs.

To allow communication, Inter-VLAN Routing must be configured using:

- Router-on-a-Stick, or
- Layer 3 Switch

After configuring inter-VLAN routing, devices in different VLANs can communicate successfully.